```
#include <stdio.h>

#include <stdlib.h>


Int main()

{

   Int a[20], n, head, size, I, j, temp;

   Int movement = 0;


   Printf("Enter number of requests: ");

   Scanf("%d", &n);


   Printf("Enter request queue:\n");

   For(I = 0; I < n; i++)

      Scanf("%d", &a[i]);


   Printf("Enter initial head position: ");

   Scanf("%d", &head);


   Printf("Enter disk size: ");

   Scanf("%d", &size);


   /* Sort requests */

   For(I = 0; I < n-1; i++)

      For(j = i+1; j < n; j++)

         If(a[i] > a[j])

         {
```

```
        Temp = a[i];
        A[i] = a[j];
        A[j] = temp;
      }

Printf("\nHead Movement:\n");

/* Move towards higher cylinders */
For(I = 0; I < n; i++)
{
  If(a[i] >= head)
  {
    Movement += abs(head – a[i]);
    Printf("%d -> %d\n", head, a[i]);
    Head = a[i];
  }
}

/* Move to end of disk */
Movement += abs(head – (size – 1));
Printf("%d -> %d\n", head, size – 1);
Head = size – 1;

/* Reverse direction */
For(I = n – 1; I >= 0; i--)
{
```

```
        If(a[i] < head)
        {
            Movement += abs(head – a[i]);
            Printf("%d -> %d\n", head, a[i]);
            Head = a[i];
        }
    }

    Printf("\nTotal Head Movement = %d cylinders\n", movement);

    Return 0;
}
```